KAIZEN MATHS

Lesson starter

Hyperbolic Functions

Hyperbolic Equations

Name _____

S1

Solve each equation exactly.

Solve sinh x = 4.

S2

Solve each equation exactly, giving both real solutions where appropriate.

Solve cosh x = 5.

S3

Solve each equation exactly.

Solve tanh x = 1 3 .

S4

Use the cosh x + sinh x and cosh x - sinh x exponential identities.

Solve 4cosh x + 4sinh x = 20.

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Hyperbolic Functions

Hyperbolic Equations

Name _____

E1

Solve each equation exactly.

Solve $\sinh x = 2$.

E2

Solve each equation exactly, giving both real solutions where appropriate.

Solve $\cosh x = 4$.

E3

Solve each equation exactly.

Solve $\tanh x = 3\,5$.

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Solve each equation exactly.

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Hyperbolic Functions

Hyperbolic Equations

Name _____

E1

Solve each equation exactly.

Solve sinh x = 2.

E2

Solve each equation exactly, giving both real solutions where appropriate.

Solve cosh x = 4.

E3

Solve each equation exactly.

Solve tanh x = 3 5 .

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Teacher answers & working

Hyperbolic Functions · Hyperbolic Equations

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S1

Solve each equation exactly.

Solve sinh x = 4.

Answer:

$x = \ln(4 + \sqrt{17})$

Working:

Use sinh x = $\frac{e^x - e^{-x}}{2}$.

$\frac{e^x - e^{-x}}{2} = 4$.

Multiply by 2: $e^x - e^{-x} = 8$.

Let $y = e^x$. Then $e^{-x} = \frac{1}{y}$.

$y - \frac{1}{y} = 8$.

Multiply by y: $y^2 - 8y - 1 = 0$.

Solve the quadratic: $y = 4 + \sqrt{17}$, since $y = e^x$ must be positive.

Therefore $x = \ln(4 + \sqrt{17})$.

S2

Solve each equation exactly, giving both real solutions where appropriate.

Solve cosh x = 5.

Answer:

$x = \pm\ln(5 + \sqrt{24})$

Working:

For a > 1, cosh x = 5 has two real solutions.

Use arcosh a = $\ln(a + \sqrt{(a^2 - 1)})$.

Here a = 5.

arcosh(5) = $\ln(5 + \sqrt{(25 - 1)}) = \ln(5 + \sqrt{24})$.

Because cosh x is even, the solutions are $x = \pm\ln(5 + \sqrt{24})$.

Teacher answers & working

Hyperbolic Functions · Hyperbolic Equations

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S3

Solve each equation exactly.

Solve tanh x = 1 3 .

Answer:

$x = \frac{1}{2}\ln(2)$

Working:

Use artanh k = $\frac{1}{2}\ln(\frac{1+k}{1-k})$.

Here k = $\frac{1}{3}$.

$x = \frac{1}{2}\ln(\frac{1+\frac{1}{3}}{1-\frac{1}{3}})$.

Simplify inside the logarithm: $\frac{1+\frac{1}{3}}{1-\frac{1}{3}} = \frac{4}{2}$.

Therefore x = $\frac{1}{2}\ln(2)$.

S4

Use the cosh x + sinh x and cosh x - sinh x exponential identities.

Solve 4cosh x + 4sinh x = 20.

Answer:

$x = \ln 5$

Working:

Factor the left-hand side: 4cosh x + 4sinh x = 4(cosh x + sinh x).

Use cosh x + sinh x = e^x.

$4e^x = 20$.

$e^x = 5$.

Take natural logarithms: x = ln 5.

Therefore x = ln 5.

Teacher answers & working

Hyperbolic Functions · Hyperbolic Equations

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E1

Solve each equation exactly.

Solve sinh x = 2.

Answer:

$x = \ln(2 + \sqrt{5})$

Working:

Use sinh x = $\frac{e^x - e^{-x}}{2}$.

$\frac{e^x - e^{-x}}{2} = 2$.

Multiply by 2: $e^x - e^{-x} = 4$.

Let $y = e^x$. Then $e^{-x} = \frac{1}{y}$.

$y - \frac{1}{y} = 4$.

Multiply by y: $y^2 - 4y - 1 = 0$.

Solve the quadratic: $y = 2 + \sqrt{5}$, since $y = e^x$ must be positive.

Therefore $x = \ln(2 + \sqrt{5})$.

E2

Solve each equation exactly, giving both real solutions where appropriate.

Solve cosh x = 4.

Answer:

$x = \pm\ln(4 + \sqrt{15})$

Working:

For a > 1, cosh x = 4 has two real solutions.

Use arcosh a = $\ln(a + \sqrt{(a^2 - 1)})$.

Here a = 4.

arcosh(4) = $\ln(4 + \sqrt{(16 - 1)}) = \ln(4 + \sqrt{15})$.

Because cosh x is even, the solutions are $x = \pm\ln(4 + \sqrt{15})$.

Teacher answers & working

Hyperbolic Functions · Hyperbolic Equations

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E3

Solve each equation exactly.

Solve tanh x = 3 5 .

Answer:

$x = \frac{1}{2}\ln(4)$

Working:

Use artanh k = $\frac{1}{2}\ln(\frac{1+k}{1-k})$.

Here k = $\frac{3}{5}$.

$x = \frac{1}{2}\ln(\frac{1+\frac{3}{5}}{1-\frac{3}{5}})$.

Simplify inside the logarithm: $\frac{1+\frac{3}{5}}{1-\frac{3}{5}} = \frac{8}{2}$.

Therefore x = $\frac{1}{2}\ln(4)$.